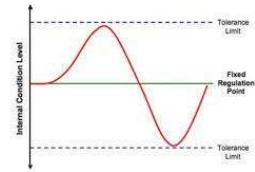


Module 8: Non-infectious Disease and Disorders

Homeostasis - How is an organism's internal environment maintained in response to a changing external environment?

Homeostasis - maintenance of a relatively stable internal environment even when the external environment is changing.



Maintenance of homeostasis involves 2 steps:

1. **Detecting stage** - involves receptors that detect stimuli
2. **Counteracting stage** - involves effectors that respond to stimuli and reduce its effects

Stimuli - any internal or external factor/change which can be detected by our body receptors.

E.g. - increase or decrease in body temperature

- Loud sound
- low/high blood sugar levels
- Bright light

Receptors - sensory nerve cells which detect stimuli

Receptor	Stimulus	Organ Examples
Thermoreceptor	Heat	Skin, hypothalamus
Chemoreceptor	Chemicals (ph, CO ₂ , O ₂)	Olfactory, tongue
Photoreceptor	Light	Retina
Mechanoreceptor	Movement	Skin, ears

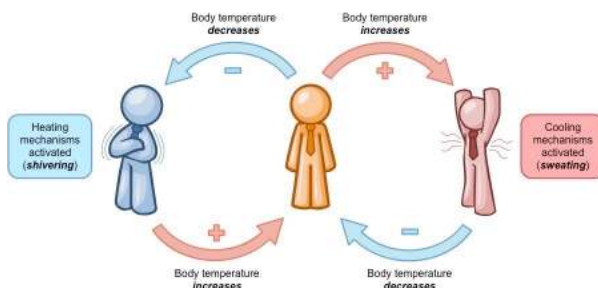
Integrator (control centre) - hypothalamus & pancreas

Effectors - **muscles or glands** that carry out a response to the stimulus -> return body back to homeostasis / reverse the initial effect of the stimulus

E.g. Iris in your eye shrinks the pupil if light is bright

Negative feedback loop - mechanisms which detect the change and induce a response which counteracts the change and returns the body to homeostasis / stable state.

Detect change -> process change -> decide on response -> returning back to set value



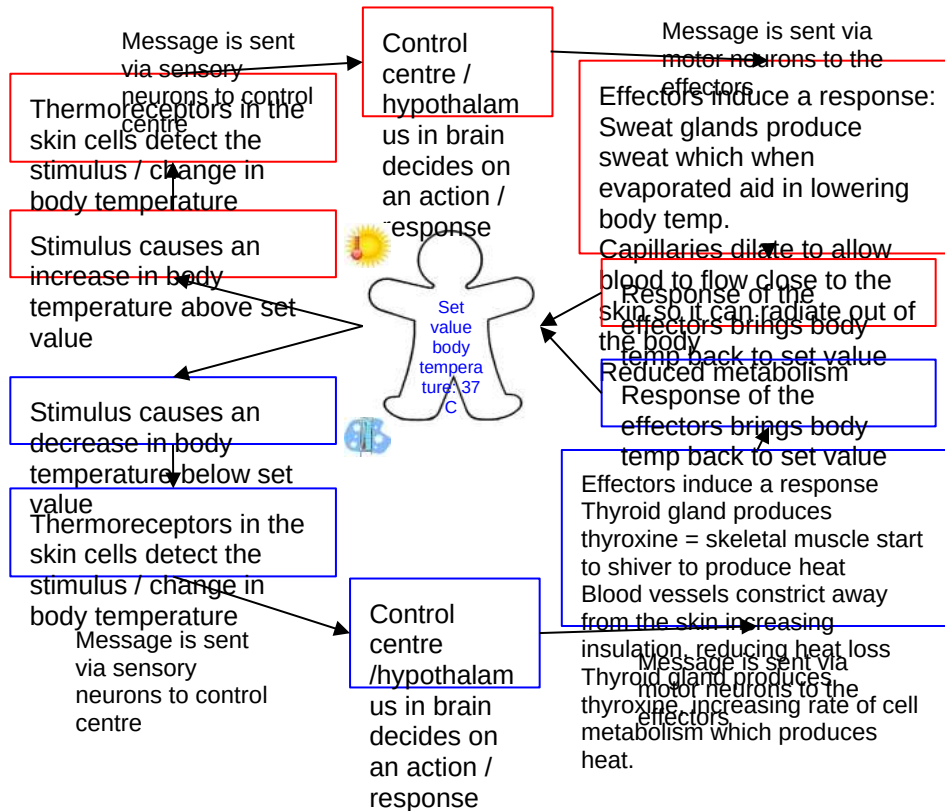
Positive feedback loop - mechanisms which

detect the change and induce a response which causes additional change in the same direction.

E.g. The production of oxytocin results in the production of more oxytocin.

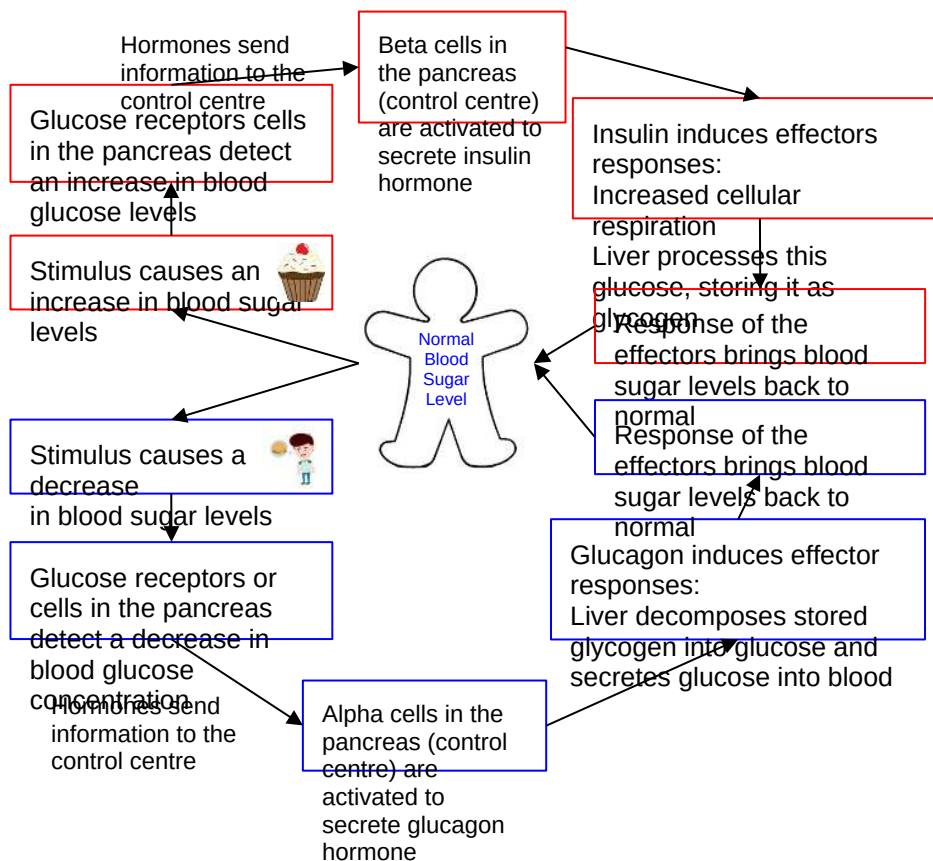
- construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited to:

- temperature (nervous system) without maintaining internal body temperature, bodily reactions do not work efficiently as enzymes denature or slow down in performing reactions



- Glucose (endocrine

system) without maintaining homeostasis, symptoms of diabetes (hypoglycemia or hyperglycemia) could occur.



- investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including:

- trends and patterns in behavioural,

structural and physiological adaptations in endotherms that assist in maintaining homeostasis

Endotherms (warm blooded) - Organisms that use internal processes to regulate body temperature in a narrow range, irrespective of the external temperature. They use the heat produced from their metabolism to help maintain their body temperatures. E.g. mammals and birds.

Ectotherms (cold blooded)- organisms that have approximately the same body temperature as the ambient temperature. They have a limited ability to maintain their body temperature at the one level, as it fluctuates according to the surrounding environment. E.g. plants, invertebrates, amphibians, reptiles and fish.

Behavioural - how can an organism's actions ensure homeostasis is maintained

Structural - How can an organism's physical features maintain homeostasis

Physiological - How can internal metabolic processes ensure homeostasis is maintained

Adaptation to maintain body temperature	Red Kangaroo (heat)	Penguins (cold)
Behavioural (actions)	- Kangaroos crouch into smaller positions so less heat is absorbed from the sun. - Kangaroos sit in the shade to reduce exposure to sun and heat absorption. - Kangaroos lick their forearms and chest to increase evaporation from the skin in order to increase heat loss	- Emperor penguins form large huddles. Huddles allow them to share body warmth, and shelters many of the penguins from the wind. Huddling can reduce heat loss by up to 50%.
Structural (physical features)	- Fur can reflect 30% of heat from the sun - Large ears of the kangaroo allow heat to radiate from blood vessels close to the skin	- Thick windproof and waterproof feathers reduce heat loss from convection occurring in the water and snow storms - The dark colored feathers of a penguin's dorsal surface (their back) absorb heat from the sun
Physiological (metabolic processes)	- Has exposed areas of skin on the forelegs to allow heat loss through convection and evaporation when licking these areas - Blood can be shunt from the tail to exposed areas of the skin on the forelegs to increase heat loss - Can decrease their metabolic rate to decrease heat production	- <i>Counter Current Change</i> Closely aligned veins and arteries allow the recycle of heat to maintain internal body temperature. Warm blood from the heart, warms the cool blood coming from the extremities. This reduces metabolism required to heat the body up as heat is conserved instead of very cold blood cooling the body temperature as it travels back to the heart. - Increased rate of metabolism

– internal coordination systems that allow homeostasis to be maintained, including hormones and neural pathways

Endocrine system and the nervous system act together to maintain homeostasis to coordinate body processes.

Endocrine System

Nervous System

The endocrine system consists of many different hormones and the glands that produce them.

1. Hormones are chemical messages that travel through the blood and lymphatic system to target organs or tissues
2. Hormones find their specific target cells which have receptors on their surface and bind to the receptor, triggering effector response

Hypothalamus - Releases many different hormones and stimulates other glands. Links the nervous system to the endocrine system. (part of both systems)

Pituitary gland - produces hormones which help regulate important functions, such as growth, blood pressure and reproduction

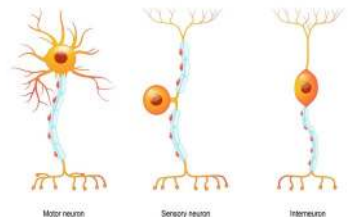
The nervous system consists of the network of nerves in the body, which are connected to the brain and spinal cord. It sends electrical impulses throughout the body through nerves. The nervous system detects changes using specialised receptor cells which generate nerve impulses which travel along sensory neurons.

Central nervous system (CNS) - brain + spinal cord

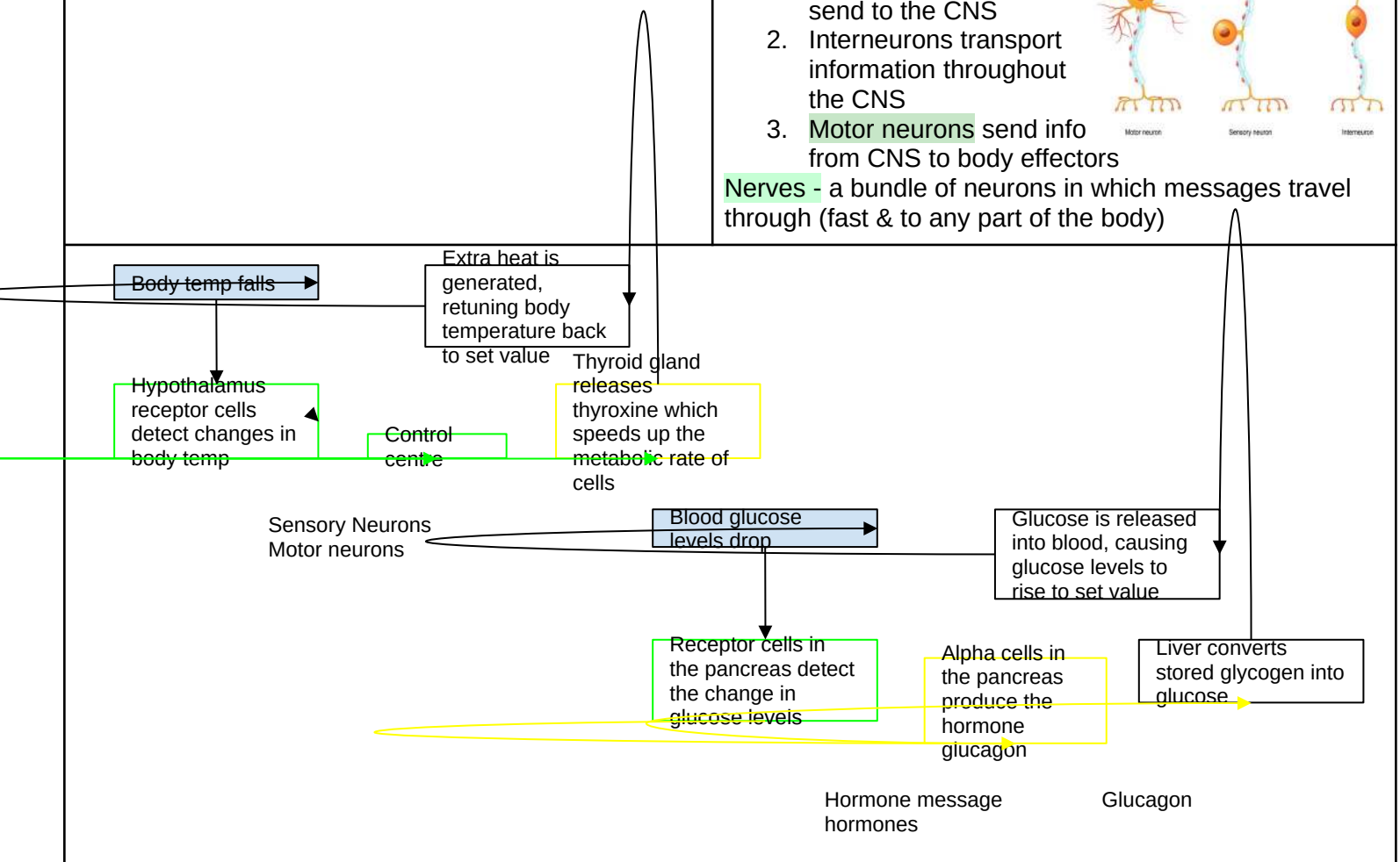
Peripheral nervous system - nerves leading to and from the body (ascending and descending nerves)

Neurons - single cells in the nervous system

1. **Sensory neurons** take info from the nerves and send to the CNS
2. Interneurons transport information throughout the CNS
3. **Motor neurons** send info from CNS to body effectors



Nerves - a bundle of neurons in which messages travel through (fast & to any part of the body)



– mechanisms in plants that allow water balance to be maintained

Plants must regulate their water salt balance as plants need water to photosynthesise. They maintain osmotic balance by closing or opening the stomate.

Stress due to lack of water

1. Cells produce abscisic acid
2. Acid is a stimulus detected by guard cells

- Guard cells move potassium out of the vacuoles
- Water leaves the guard cells due to osmosis
- Guard cells close, reducing evaporative loss of water from stomate opening

Too much water

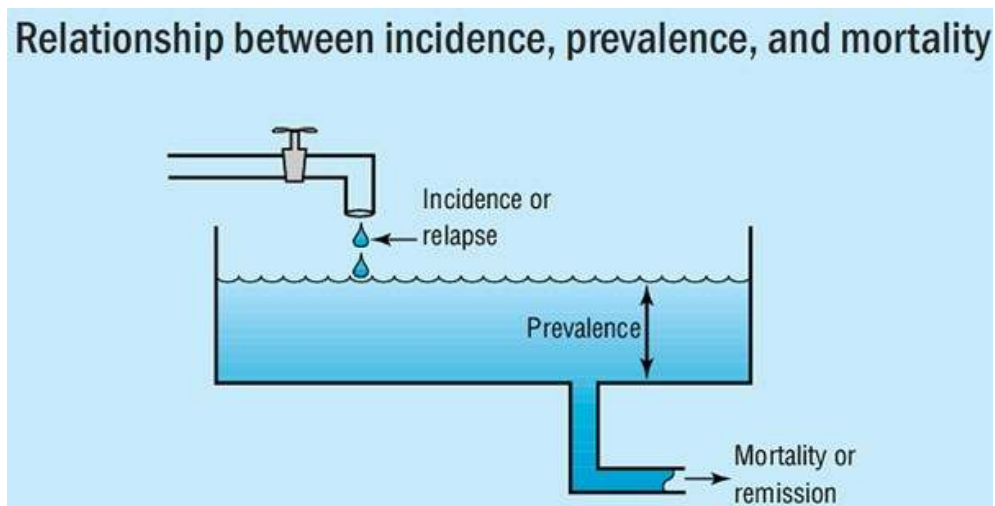
- Potassium ions move into vacuole
- Water follows the potassium ions into the vacuole due to osmosis
- Guard cells expand and swell causing them to open
- An opened guard cell allows water to evaporate from the plant through opening the stomate

Other mechanisms / adaptations

- Orientation of leaves** e.g away from sunlight or vertically hanging/ curling of leaves to reduce sun exposure, which will lead to less water loss. (stomata on inner side of curl, increasing humidity around stomata, reducing rate of water evaporation)
- Reduced leaf size to reduce the amount of water lost through transpiration
- Fleshy stems and leaves swell to store more water
- Waxy cuticles** prevent evaporation
- Sunken stomata

E.g. the Eucalyptus tree has vertically hanging leaves, fewer stomata, waxy cuticles

Causes and Effects - Do non-infectious diseases cause more deaths than infectious diseases?



- investigate the causes and effects of non-infectious diseases in humans, including but not limited to:

Type of non-infectious disease and definition	Specific Example	Cause & Effect	Host Responses (includes Effect / symptoms)
genetic diseases Genetic diseases are hereditary diseases that are passed down through gametes fusing.	Cystic Fibrosis	- Inherited disease caused by a mutation of the cystic fibrosis transmembrane conductor regulator gene on chromosome 7 - Changes the protein that regulates the normal movement of sodium chloride in and out of cells. The chloride ions become trapped inside the cell, disturbing the osmosis cycle as the	- Disrupts functioning of several organs including the lungs, liver, reproductive tract as they become clogged with thick mucus = chest infections, liver failure and excessive salt in sweat - Increased coughing → attempt to expel the mucous - inflammation response to the

		water is no longer attracted to space outside the cell, causing mucus to thicken along the respiratory and digestive track - Cilia are also flattened due to the weight of mucous and are unable to move bacteria out of the airway = thick and sticky mucus	amount of bacterial pathogens trapped and scarring on lung tissue from excessive coughing - Bacterial response from macrophages - Acquired immune response - antibody production + cytotoxic T cells reduce pathogen infection
- diseases caused by environmental exposure	Melanoma - Skin Cancer Asthma Asbestosis	Melanoma - Radiation from the sun promotes the fusion of thymine bases to create thymine dimers - Thymine dimers cause a bulge in the DNA and can change or disrupt DNA expression, preventing normal transcription and replication which can result in cancerous cells	- Enzymes try to cut out the thymers during DNA replication and other, however if the enzymes aren't enough \ there's a lack of enzymes it can lead to the duplication of cancerous cells - Natural killer cells – (WBC) – release cytotoxic protein perforin which forms pores in the cell membrane causing fluid to rush in and the cell to burst.
- nutritional diseases Nutritional diseases are any illnesses that occur due to poor diet choices or genetic factors.	Anemia Iron Deficiency Obesity Diabetes Lactose Intolerance	Lactose Intolerance - The body is unable to digest lactose, causing stomach pains, gas, bloating and diarrhea - People with lactose intolerance don't make enough of the enzyme lactase, which is needed to digest lactose.	Gas / bloating / diarrhea - Which leads to → diarrhea to help the host get rid of the lactose that cannot be broken down
- cancer Cancer is a disease of the cells. It encompasses a large number of diseases characterised by the development of abnormal cells that divide uncontrollably. These abnormal cells can spread and destroy other healthy tissue, causing damage to vital organs.	Breast Cancer Melanoma Lung Cancer	Breast Cancer - Breast cells mutate in cell division. Breast cells divide more rapidly causing a lump known as a tumor - Normal cells age, die and are replaced by new cells. A tumor occurs due to something disrupting this cycle such as mutations in the BRCA1 or 2 gene	Granulomas - White blood cells surround the cancerous cells to stop them from spreading and growing Natural Killer Cells - Cancerous cells often lose their antigens that allow the body to identify the cell as its own. As a result, the cancer cells become vulnerable to killer cells

Non-infectious diseases have exceeded infectious diseases as the leading cause of death and disability worldwide. It contributes to 73% of all deaths and 60% of the global burden of disease. 70% of all deaths due to diseases occur in developing countries due to a lack of health services, lower socioeconomic health barriers and infrastructure.

- represent data to show the incidence, prevalence and mortality rates of non-infectious diseases +
- analyse patterns of non-infectious diseases in populations, including their incidence and prevalence, including but not limited to:

Prevalence: the number of cases of disease in a population at a specific time

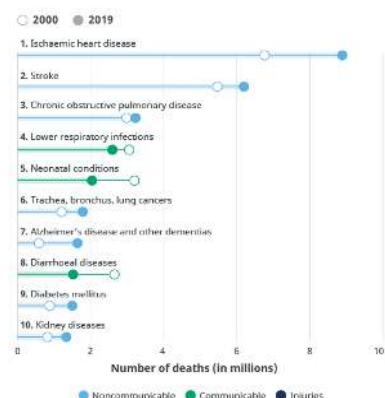
Incidence: the number of new cases of disease occurring in a population

Mortality Rates: the amount of deaths

WHO - World Health Organisation is a statistically reliable source of information (trustworthy up to date information) and is reliable as it correlates with other websites.

Non-infectious diseases have risen in prevalence due to lifestyle factors such as environmental exposure and nutrition.

Leading causes of death globally



– nutritional diseases

Diabetes (type 2)

Incidence & Prevalence	Mortality Rates	Analysis of patterns & trends
- Has increased by 60% since 2000 and is still on the rise due to poor lifestyle behaviours - 85% of diabetes have type 2 - Diabetes grew in prevalence after the age of 45	- Australia's 7th leading cause of death - Relatively unchanged	- High rates of type two diabetes indicates prevalence is due to poor lifestyle choices when it comes to diet and exercise = obesity & the ageing population - Older people are more susceptible to diabetes due to decreased insulin production and increased insulin resistance

Coronary Heart Disease (stroke)

Incidence & Prevalence	Mortality Rates	Analysis of patterns & trends
- Incidence has increased by 120% in the last 30 years - Prevalence increases with age (blood vessels become inelastic and heart weakens)	- Mortality rates have decreased due to tech innovation and development and health campaigns - Leading cause of mortality	- Lifestyle factors such as diet (high in sodium and fat and low in fruit/veg) and less people exercising has contributed to increased prevalence of CHD - Higher mortality rates in lower socioeconomic, ATSI and rural groups due to less access to nutritional food and health services

– diseases caused by environmental exposure

Melanoma

Incidence & Prevalence	Mortality Rates	Analysis of patterns & trends
- Incidence increases with age (cumulative exposure) - Incidence has increased by 60% (cumulative disease, therefore trends may be reflective of older generations pre-campaigns) - More men are diagnosed than females (females more inclined to wear sunscreen) - Risen due to an ageing population	- Mortality rates have been increasing due to increased incidence rates - 10th leading cause of cancer death	- Amongst the younger age groups Incidence and mortality rates have fallen as a % of the group due to government funded health promotions such as slip slop slap - Greater prevalence in rural Australia and around the coast due to nature of work (lower prevalence ATSI) - Greater prevalence in Australia due to our location under a hole in the ozone layer and therefore greater exposure to strong UV rays

Lung Cancer

Incidence & Prevalence	Mortality Rates	Analysis of patterns & trends
- 5th most common cancer diagnosed in Au - Incidence rates increase with age - Incidence for females has risen while fallen for men (little change to average)	- Leading cause of cancer death - Mortality rates on the rise but likely due to lung cancer being exposure related and therefore rates linked to pre-campaign implementation	- Due to cumulative effect of nicotine, people that smoke for longer and people that smoke in greater quantities have a higher risk of lung cancer - High prevalence amongst lower socioeconomic groups and rural populations due to greater rates of smoking

Epidemiology - Why are epidemiological studies used?

Epidemiology: the study of connecting trends such as distribution, prevalence and determinants of health to the cause of a disease & patterns.

Proves correlation of risk factors to a disease trends

Epidemiology studies are essential to help better understand the cause of a disease, the efficacy of different treatments and prevention methods.

Epidemiological Studies		
Descriptive / Ecological Study	Analytical	Intervention / Experimental
Studies involve researching patterns of the disease by asking who, what, when, where it is prevalent. - Prevalence + incidence - Location - Sections of a population (age, gender, occupation, socioeconomic status)	More data is collected to test the hypothesis by conducting research that measures the association between a particular factor/cause and a disease. Case Control Studies - looks at people already with disease Compares people with the disease to people without the disease and looks for differences in exposure to be the possible cause of a disease	Studies used to test the effectiveness of treatment or public health campaigns. Experimental Involves randomly placing people suffering from the disease into two groups. One group receives the trial

- Length of infection <u>Data collected is then used to form a hypothesis about the cause of the disease.</u> Advantages: snapshot of population, prevalence of disease and risk taking behaviours, correlates disease with risk factors that may be unknown Disadvantages: costly to do a wide census, only specific in a point in time, unclear if risk factor cause disease or disease caused increase in risk factor	Advantages: inexpensive, smaller sample groups, less time consuming Disadvantages: difficult to select a control group who are a good representative of the entire population, inefficient for rare exposures, can only study one outcome at a time Cohort Studies - compare people who have the risk factor vs people who don't and observe who develops the disease Two groups of similar people (age and risk of disease) who are free of the disease are studied. One group is exposed to the possible cause and the other is not. These groups are followed over a long period of time to compare the resulting incidence of the disease. Advantages: shows exposure and outcome link, calculates incidences and prevalence, can study multiple diseases Disadvantages: costs a lot, takes time for study to be completed, people may drop out of study, large sample size required	drug and the other receives a placebo. Follow each group for over 20 years and analyse if there is a difference in outcomes. Advantages: identifies a causal relationship, variables can be minimised, Disadvantages: expensive, take a long time, no certainty if cohort is a good representation of populace, ethical concerns
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- investigate the treatment/management, and possible future directions for further research, of a non-infectious disease using an example from one of the non-infectious diseases categories listed above

Melanoma (environmental exposure)

Treatment/Management Treatment aims to remove cancer and reduce the risk of cancer spreading or coming back.	Future Directions Aims to further prevent future incidence. This is done through better educational programs and better development of treatment and management
Surgery is used to remove the melanoma lump (tumor) before it is able to metastasise to other parts of the body.	Educational programs can be used to increase awareness of the effects of unsafe sun behaviour. This will increase people protecting their skin, reducing the prevalence of melanoma.
Immunotherapy is a drug treatment to help the immune system fight melanoma. The body's immune system might not attack cancer because the cancer cells produce proteins that help them hide from the immune system cells. Immunotherapy works by interfering with that process.	Genetic Programs Some individuals are more susceptible to developing melanoma due to genetics. Two genes have been primarily linked to familial melanoma; they are called CDKN2A and CDK4. Programs can involve DNA sequencing individuals to identify their future risks.
Radiation involves high-powered energy beams, such as X-rays and protons, to kill cancer cells. Can be used in addition to surgery.	Immunotherapy Tag cancer cells so that the immune system can identify
Targeted Therapy involves the use of drugs that affect the molecules that control the growth of the tumor cells, which effectively stop the growth and spread of the tumor to	Vaccines Use a vaccine to expose the body to cancer 'antigen'.

- evaluate the method used in an example of an epidemiological study

Lung Cancer Epidemiological Study - a study conducted in Queensland to test the hypothesis that the prevalence of skin cancer is greater in outdoor workers than indoor workers.

Control: indoor workers

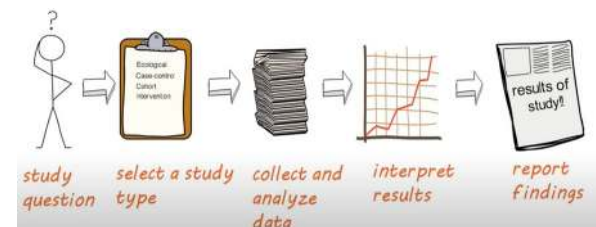
- Done in this specific town where the prevalence was already high
- Picked people randomly from the electoral roll to ensure a broad range of lifestyles are included
- Flaws: people who participate may be of lower economic groups as they may be unemployed or people of older age as they have more time or people already concerned about skin cancer
- Tried to compensate flaws by following up non-responders
- Participants were examined by epidemiologist
- Variables that may have influenced the results: age, previous occupation, complexion, freckling, ethnicity, recreational habit, diet, smoking, alcohol, drugs, other diseases, genetics

No statistically significant correlation between outdoor occupations and incidence of skin cancer. It however did show correlation between skin cancer and complexion, freckling on the back and cutaneous sun damage. Results may have been due to outdoor workers more likely wearing hats and sunscreen.

Evaluation: the study did not support the hypothesis but it was still valuable as it established a correlation between fair skin and damaged skin to high incidences.

How to Evaluate / Create an epidemiological study

- Was the aim of the study achieved ? e.g. cause of disease, correlation in trends, management strategies tested for effectiveness. **Is the data answering the aim**
- Hypothesis (....more likely to Because)
- **Was the appropriate type of epidemiological study used (Descriptive, Analytical - case control or cohort, Experiment)**
- **Reliability:** results were consistent, matched with other studies, peer reviewed, **variables are controlled**, accurate forms of measures, trained professional collected the data
- **Validity:**
- **conducted over a long period of time**
- **very large sample group (10,000 +)**
- **Collect a range of relevant data from a large group of both affected and unaffected people** (set questionnaires)
- Have participants that represent a broad range of society and lifestyles (participants should not be eliminated on the basis of their answers as this reduces validity)
- **control group (show that independent variable is the cause of the change)**
- Data on incidence, prevalence, mortality, morbidity rates
- Identify the possible cause of the disease (determinants of health) and risk factors
- Result in the development of a management plan with strategies to control disease
- Evaluate the effectiveness of control and treatment programs
- Final data should be peer reviewed
- **describe what you would do to analyse this data**



- evaluate, using examples, the benefits of engaging in an epidemiological study

Why is epidemiology useful?

- Benchmark the prevalence of a disease and monitoring any changes overtime
- Guides public health warnings and educational campaigns to be more specific
E.g. lung cancer campaigns target smokers, melanoma campaigns are about sun safety
- Informs public policy and law making
- Finds patterns to identify individuals more at risk
E.g Women aged between 50-74 are advised to regularly breast screen
Is it a cumulative effect?
- Evaluates effectiveness of treatment to inform where government spending is required for further research, development and health services (saves money and time)

E.g.

Diabetes - canteens selling healthier food, compulsory PE till year 10 (exercise and diet)

Alcohol and Smoking - Healthy Harold education program to influence children in influencing healthy habits

Smoking - no smoking on public transport and public parks and workplaces

Prevention - How can non-infectious diseases be prevented?

- use secondary sources to evaluate the effectiveness of current disease-prevention methods and develop strategies for the prevention of a non-infectious disease, including but not limited to:
 - educational programs and campaigns

Melanoma - Slip, Slop, Slop: Was developed by the Cancer Council in Australia. Increased awareness of the link UV exposure & sunbed tanning to skin cancer. It also informed the public of preventive behaviours and habits that will reduce the risk of developing skin cancer. Programs targeted children and young adults but also adults with children. Has changed over the years to also include seek (seeking shade) and slide (sliding on sunglasses) after further research on the harms of sun. Some campaigns also changed to emphasis on changing behaviours by showing the consequences as many people chose to ignore campaigns.

Advantages:

- Melanoma incidences have fallen, preventing tens of thousands of cases
- Reaches a large population through multiple platforms
- Ensures early detection and prevents incidence
- Inspires local campaigns which are better suited to the demographic e.g. no hat no play in schools
- Tailored to specific target groups e.g. Campaigns are more effective and harder to ignore
E.g. Slip, slop, slap for young children, anti-tanning beds for young women

Limitations:

- Campaigns that target children don't impact parents who have control over children's lifestyles
- Campaigns don't enforce the change and therefore people ignore the information
- Some people may find it hard to stop, don't have the will power e.g. smoking addicts

Coronary Heart Disease - Jump Rope For Heart or Healthy Canteens

- genetic engineering

Preimplantation genetic Testing for deafness: eggs are harvested and collected and then fertilized. Each embryo is then tested for the mutated gene. Embryos that were free of the gene or were carriers were retained and the other embryos were destroyed. The retained embryos are implanted and the offspring born will not develop the genetic disease.

Advantages

- More ethical than testing and aborting foetus
- Cost saving - family and health system don't have to pay for treatments in the future
- Carriers can be identified and discarded - remove disease from future family generations
- High accuracy

Limitations

- Some people are culturally against IVF
- Discarding embryos is unethical

Genetically Modified Organisms for nutritional diseases: golden rice is a genetically modified rice species that contains genes from maize and genes from soil bacterium. This allows the rice to produce beta-carotene, which is used by the body to produce Vitamin A. This prevents blindness and a compromised immune system.

Advantages

- Provides people in developing countries with a more nutritious diet for a cheap price
- Rice is a staple food in many countries and is easily incorporated into the everyday diet
- Reduces the risk of blindness (caused by lack of vitamin A)
- Seeds are easily replanted

Limitations

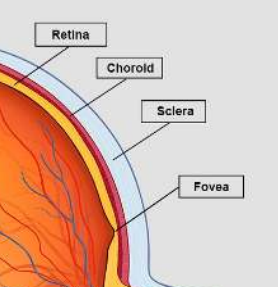
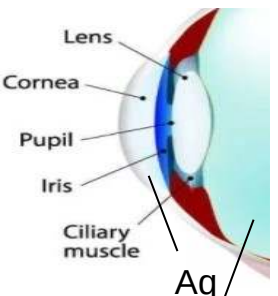
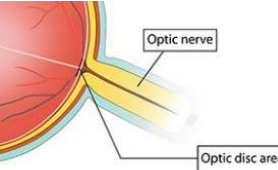
- smaller yield being produced compared to normal rice
- Opposition to genetically modified organisms due to culture, ethics, religion
- Can cost more which third world countries cannot afford to produce

Technologies and Disorders - How can technologies be used to assist people who experience disorders?

- explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example:

- Visual disorders

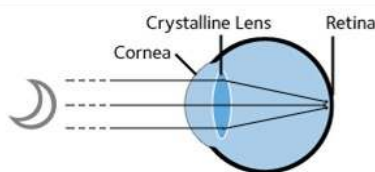
The eye contains **photoreceptors** that receive light energy and convert it to nerve impulses (electrical energy) sent via the **optic nerves** to the brain where they are interpreted as images.

Parts of the Eye	Definition	
	Sclera	The tough white outer layer of the eye that protects the eyeball and gives it shape
	Choroid	Layer of blood vessels between the retina and sclera
	Retina	The inner lining of the eyeball which contains photoreceptors which detect light and convert light signals to electrochemical signals which can be interpreted by the brain
	Fovea	Region of the Retina where the greatest focusing occurs
	Conjunctiva	mucus membrane that covers the front of the eye and eyelids to lubricate and avoid friction when blinking.
	Cornea	Dome-shaped casement that covers the front of the eye and is responsible for refracting 2/3 of light , allowing focus
	Aqueous Humor	Fluid which is found between the cornea and the lens, filling the front of the eye & nourishing the eye
	Lens	A transparent bi-convex lens which refracts light to achieve focus
	Ciliary Muscles	Controls the tension of suspensory ligaments and therefore the thickness of the lens.
	Suspensory Ligaments	Ligaments that attach the ciliary muscles to the lens, allowing the lens to change shape.
	Pupil	Gap in the iris which allows light to go through to the retina
	Iris	Coloured muscles which controls the size of the pupil and therefore the amount of light entering the eye
	Vitreous Humor	Gelatinous substances which fills the eyeball between the lens and retina
	Optical nerve	Transmits electrical (visual) signals from the retina to the visual cortex in the brain
	Optic disk	Also known as the blindspot; is where there are no photoreceptors (as this is where optic nerves join) so the brain fills in the missing parts to create a whole image

Accommodation: the ability for your eyes to change focus so that objects can be seen sharply at varying distances. When the lens of the eye alters (thickens, stretches) to make an image more clear, it is said to be accommodated.

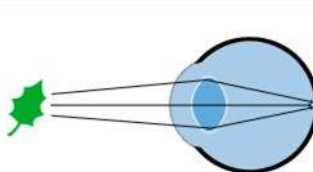
Far Object: thin bi-convex lens

Ciliary Muscles relax, causing tension of ligaments. causing lens to flatten



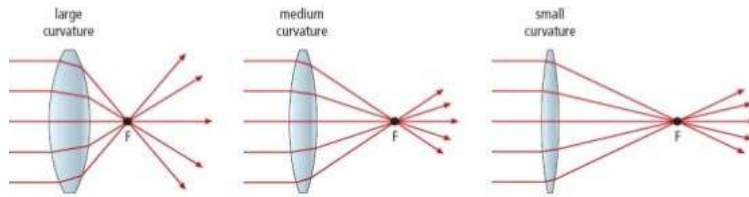
Close Object: thick bi-convex lens

Ciliary Muscles contract, releasing tension of ligaments. causing lens to become spherical



Light coming from distant objects is almost parallel

Refraction: bending of light as it passes from one medium to another of different density



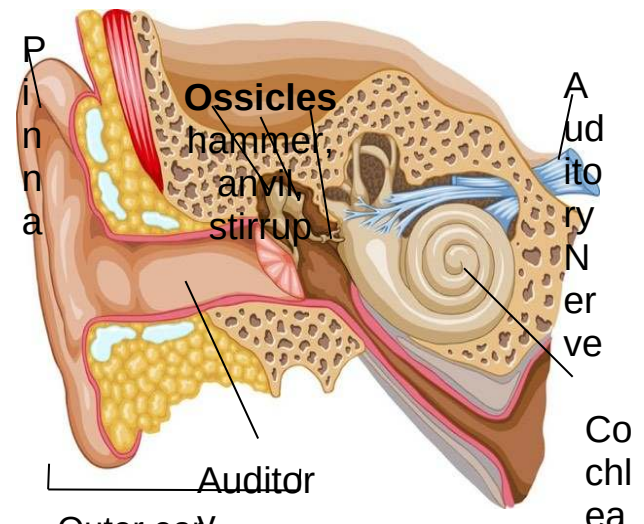
- The thicker the lens, the closer the object. This is because light is coming from all directions and therefore requires greater refractions.
- The thinner the lens, the further the object. This is because light coming from distant objects are closer to parallel and therefore require less refraction.

Visual Disorders

Myopia	Hyperopia
- Short sightedness (inability to focus on distant objects) - Occurs due to the light from distant objects being bent or refracted more than necessary (too curved) - Can be due to the cornea being too curved or due to a bulging cornea or elongated eyeball (image doesn't reach back as longer than normal) - As a result, images are focused in front of the retina rather than on the retinal surface = unfocused image Light is focused in front of the retina	- Farsightedness (inability to focus on close objects) - The lens of the farsighted eye can no longer form a very rounded shape required to bend light to view close objects (too flat) - As a result, images are focused at a location behind the retina meaning the light-detecting cells will perceive a blurry image - Most commonly occurs at an older age as lens becomes less flexible, making it hard for lens to become round or due to a short eyeball Light is focused behind the retina
Cataracts (progressive clouding of the lens) The lens of the eye is made up of mostly water and protein. The protein is arranged to allow light to pass freely. Sometimes protein clumps together clouding small areas of the lens, obstructing light from reaching the retina, causing blurry vision.	

- Hearing loss

1. Sound waves (sound energy) are directed into the outer ear by the **pinna** and travel through the **ear canal** until it hits the **eardrum (tympanic membrane)**.
2. When sound waves hit the tympanic membrane, they cause it to vibrate (kinetic energy).

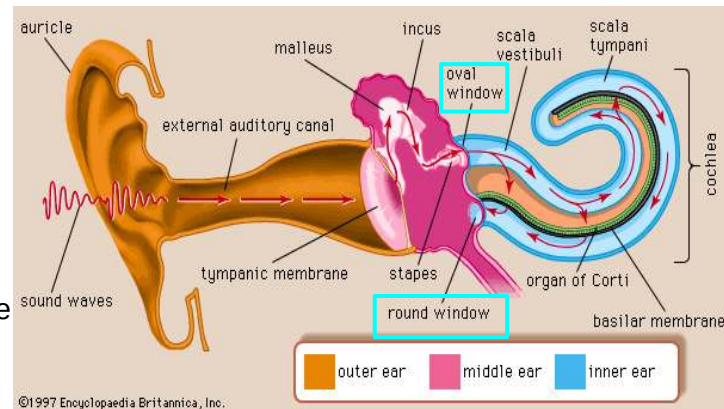


pinna + eardrum
canal + eardrum

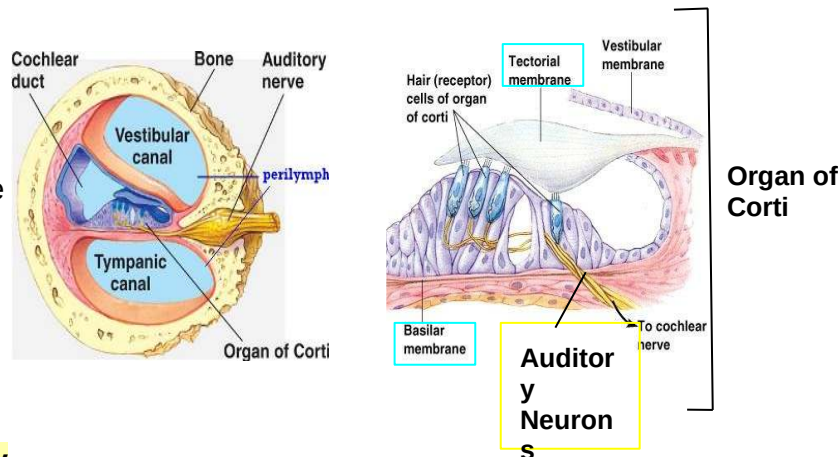
- Vibrations are transferred to tiny bones in the middle ear called the **ossicles (hammer, anvil, stirrup)**. These bones amplify and transfer kinetic energy to the oval window.
- The **oval window** is the membrane that separates the middle ear from the fluid-filled inner ear.

- Vibrations pass from the **oval window** into the **upper canal**, passing the **middle canal** through to the **lower canal** to the **round window** which acts as a valve, relieving built up pressure from the sound waves entering into the cochlea.

The Cochlea is a fluid filled organ with 3 canals (upper, middle, lower) and contains the **Organ of Corti**.



- The **middle canal** contains **receptor cells (hair cells)** located between the basilar membrane and tectorial membrane
- When **sound waves** move through the canals, it pushes up on the **basilar membrane**. This causes bending of the hair cells against the **tectorial membrane**.
- Bending of the hair cells stimulates electrical impulses which are transferred to the brain by the **auditory nerves**. (converts kinetic energy to electrochemical energy)



Sounds of different pitches are detected by different regions of the Corti.

- High pitch sounds stimulate hair cells closest to the oval window (base)
- Low pitch sounds stimulate hair cells at the other end of the cochlea (apex/centre) (this is why people normally lose their ability to hear high frequency sounds first as these are the hairs encountered in the cochlea first)

There is a finite number of hair cells and if damaged (due to ageing) frequencies corresponding to the damaged cells will not be heard.

Pinna -> ear canal -> eardrum (tympanic membrane) -> ossicles (hammer, anvil, stirrup), -> oval window -> cochlea -> round window

Hearing Disorders

(Focus on)

Conductive Hearing Loss - Bone conduction implants - Hearing aids	When something is blocking the sound waves from passing through to the inner ear . It can be caused by things like inner ear infections and trauma to the outer ear or middle ear. It can result in the loudness of sound changing.
Auditory Processing disorders	This is loss of hearing due to some type of problem with the brain processing the information coming from the auditory systems
Sensorineural	Caused by damage to the inner ear including the cochlea, hair cells and

hearing loss - Hearing aids - Cochlear implants	auditory nerves. Damage to the inner ear prevents kinetic energy of vibration being transformed into electrical impulses to be sent to the brain for interpretation. It can be either congenital hereditary, acquired through trauma, disease (tumours or meningitis) or excessive noise exposure. Damage is normally permanent and affects loudness and clarity of sound.
Mixed	Can be a combination of the above types.

Revision - Passive Transport (movement without use of energy)

Osmosis: movement of water from an area with low substance concentration to an area of high substance concentration

Diffusion: movement of particles from an area where they are of high concentrated to an area of low concentration

Active Transport

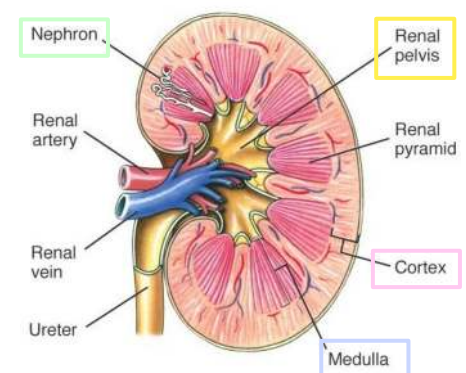
When cells use energy to transport substances from where they are less concentrated to where they are more concentrated.

– loss of kidney function

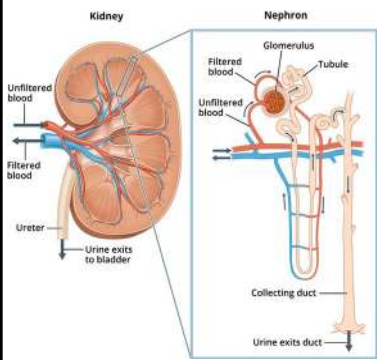
Kidney - organ that has the function of removing nitrogenous waste from the bloodstream and maintaining optimum levels of substances such as salt and water.

FUNCTION of the KIDNEY

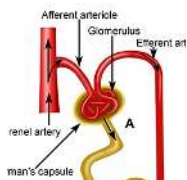
1. Excrete waste products
2. Regulate water and salt balance
3. Maintain acid balance
4. Retain substances vital to the body (glucose, proteins)

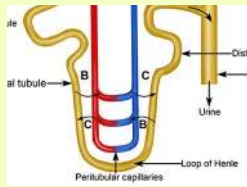
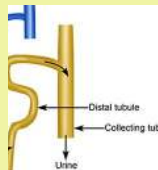


The functioning part of the kidney is called the **nephron**. The kidney contains millions of these units which extend from the **cortex** of the kidney to the **medulla**.

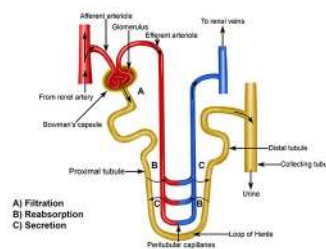
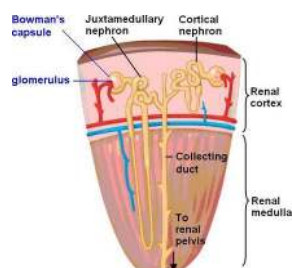
Part of the Kidney		Definition
	Cortex	Outer part of the kidney which provides space for arterioles and venules and is where filtration occurs in the nephrons.
	Medulla	Inner part of the kidney where reabsorption occurs in the nephrons.
	Pelvis	Acts as a funnel, collecting the urine released by the medulla to be sent to the bladder.
	Nephrons	Long tubes / filtering units where fluid/substances are filtered through
	Renal Artery	Carries blood to the kidney
	Renal Vein	Carries blood away from kidney

Parts of the Nephron		
	Glomerulus	A clump of capillaries that are surrounded by the Bowman's capsule
	Bowman's Capsule	Cup like sac at the beginning of the nephron which absorbs substances from the bloodstream.



	Proximal Tubule	Where two-thirds of the glomerular filtrate, including water, important ions, and small molecules, is reabsorbed.
	Loop of henle	Long U-shaped portion of the tubule which recovers water and sodium chloride from urine
	Distal tubule	End portion of the tubule which selectively secretes and absorbs different ions to maintain blood pH and electrolyte balance
	Collecting duct	Collects the urine from the nephrons.

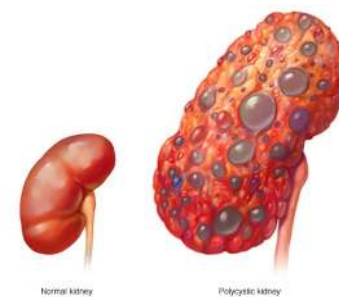
Function:



1. Filtration	2. Reabsorption	3. Secretion	4. Excretion
Moving waste materials dissolved in the fluid of the blood from the blood vessels into the nephrons to be removed from the body (passive transport)	Essential components that were filtered out of the bloodstream are returned (amino acids, glucose, minerals, vitamins) via active transport .	Occurs in the distal tubules. Moving substances actively from the blood into the nephron	The water, nitrogenous wastes and other wastes that remain in the collecting duct are known as urine.
- Blood flows into smaller blood vessels called capillaries - These capillaries form a clump called a glomerulus and is surrounded by the bowman's capsule - Substances such as nitrogenous waste, water, glucose, salt and amino acids diffuse from the capillaries into the bowman's capsule (start of the nephron) - These substances are known as glomerular filtrate	Proximal Tubule Glucose and salt moves back into the capillaries via active transport - Drugs and hydrogen move into the nephron Descending Loop of Henle - Water moves out of the nephron due to osmosis, creating a concentration gradient Ascending Loop of Henle - Salt moves out of the nephron into the medulla due to diffusion, following the osmosis of water due to a concentration gradient (Urea and other wastes are not reabsorbed)	Distal tubule - Toxic substances from the blood are actively moved into the tubules for removal e.g. urea, uric acid, ammonia, hydrogen ions, drugs - Filtrate then moves into collecting ducts which flow into the renal pelvis	Urine collects in the renal pelvis. The urine travels from the pelvis through the ureters (tube) to the bladder, after which it is eliminated from the body through the urethra. Filtrate is now called urine.

Kidney Disorders

Polycystic Kidney Disease - an inherited disorder in which clusters of cysts develop within the kidney, causing the kidney to enlarge and lose function overtime. PKD cysts can slowly replace much of the kidneys, reducing kidney function and leading to kidney failure. PKD can also result in high blood pressure which can damage blood vessels throughout the kidney. This can reduce the blood supply to the kidney, reducing efficiency of blood filtration. Nephrons become damaged due to the high pressure of



blood being pushed through the blood vessels of the glomerulus. Larger molecules may 'leak' into the tubules, causing blockages and further damage.

(PKD is a **form of chronic kidney disease (CKD)** that reduces kidney function and may lead to kidney failure)

Symptoms: fatigue, bloody or foaming urine, loss of appetite and increased risk of chronic heart disease.

Chemicals filtered out of the blood: acids, salt, urea

Reabsorbed: water, glucose

- **investigate technologies that are used to assist with the effects of a disorder, including but not limited to:**
 - **evaluate the effectiveness of a technology that is used to manage and assist with the effects of a disorder**
- hearing loss: cochlear implants, bone conduction implants, hearing aids

Type of hearing loss	Technologies to assist the disorder			
	Extent of hearing loss			
	Mild	Moderate	Severe	profound
Conductive		Bone conduction hearing devices and hearing aids		
Sensorineural	Hearing aids assistive hearing devices	Hearing aids and hybrid cochlear implants	Hearing aids and cochlear implants	Cochlear implants
Mixed hearing loss		Middle-ear implants and hearing aids	Middle-ear implants and hearing aids	Cochlear implants

Hearing technology	External ear	Middle ear	Inner ear	Cochlear nerve
Hearing aid	✓	✓	✓	✓
Bone conduction implant			✓	✓
Cochlear implant				✓

Technology	Hearing Aid	Cochlear Implant / Bionic Ear	Bone Conduction Implant
Description	Small electrical devices that sit behind the ear. They consist of a microphone, amplifier, receiver and speaker. - The hearing aids detect sound waves which is transferred to electrical energy - Electrical energy is then transformed back into sound waves to be amplified (increased in volume) into the auditory canal (when there is damaged but still functioning outer, middle or inner ear and these parts of the ear need assistance)	Sends sounds as electrical impulses directly to the auditory nerves to be sent to the brain. It returns a sense of hearing to people who have damaged middle or inner ear function. External: - Microphone detect sound waves and sends them to the speech processor - Speech processor converts sound waves into radio waves - Radio Waves transfer the signal to the receiver in the skull Internal - Radio Waves are transformed into electrical impulses which travel through electrodes - electrodes are wires that stimulate the different auditory nerves in the cochlear at different points depending on the frequency (passing damaged hair cells)	Alternative option to hearing aids. Suitable for people with conductive and mixed hearing loss, as well as those with single sided deafness. Involves a microphone which detects the sound and transforms it into vibrations. These vibrations go directly to the cochlea, where the process proceeds as normal. They therefore bypass the damaged outer and middle ear - when there is a functioning inner ear (through the bone directly to the cochlear)
Advantages	- Useful when there is damage to the outer or	- Useful when there is damage to the inner ear, allowing people to communicate	- Can help people who are physically unable to wear

	middle ear - Much cheaper than a cochlear implant with less risk involved - Easy to install	- People born deaf now have the ability to hear, preventing development delays in language, hearing and speech - Can improve the quality of life - Can connect to bluetooth	conventional hearing aids
Limitations	- Will not work if damage is done to the middle ear or auditory nerve - Hearing aids are amplifiers, making the sounds in the environment louder meaning sounds including background noise are louder	- It is different to hearing sound - Requires the person to learn to interpret sensations received which takes time to learn - Surgery has the risk of damaging facial nerves and causing infection - Must wear an external sound processor - Initial and ongoing costs for readjustments - The surgery will destroy any residual hearing	- Requires surgery - Abutment area must be kept clean - Can cost a lot

– visual disorders: spectacles, laser surgery

Spectacles

Glasses correct vision by changing the angle at which light hits the cornea, adjusting for incorrections by the cornea which causes the focal point to deviate from the norm. (can also wear contacts)

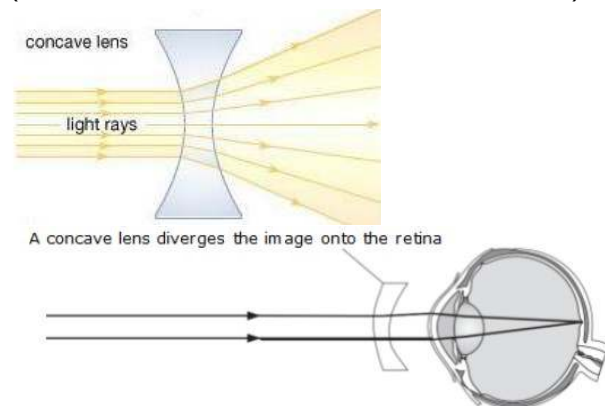
Effectiveness

- Affordable and easy to wear
- Ready made spectacles can be purchased easily and cheaply
- Contacts are difficult to apply and can dry out the eye or cause an eye infection

Myopia

- Are corrected by wearing **concave lenses**
- Causes light rays to bend outwards before they reach the eye
- Allows the focused image of a distant object to fall on the retina rather than in front of it

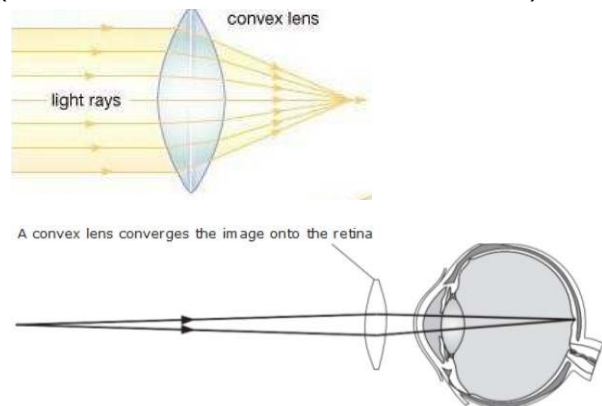
(decreases refraction so it meets the retina later)



Hyperopia

- Are corrected by wearing **convex lenses**
- Causes light rays to bend inwards before they reach the eye
- Allows focused image of a near object to fall on the retina rather than behind it

(increases refraction so it meets retina earlier)



Laser Surgery

- Vision correction that involves reshaping the cornea to correct refractive errors
- Uses cool temperature lasers
- LASIK surgery: a thin flap created in the cornea to allow it to be reshaped, the laser removed the exact amount of tissue needed to reshape the cornea, the cornea flap is then replaced

Myopia - cornea is flattened	Hypertonia - cornea is steepened	Astigmatism - cornea is smoothed
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Effectiveness

- Significantly high costs
- Have side effects including dry eyes and temporary visual disturbances
- To be successful, patients must have normal shaped corneas with adequate thickness (patients with

thin irregular shaped corneas are not suitable)

- Vision is not always 100% corrected, meaning person may still need glasses
- Not recommended for people with autoimmune disease or diabetes as both of these conditions may have damaged the cornea

Cataract Surgery

- The cloudy lens is removed and replaced with an artificial lens
- Involves making a small incision in the eye, breaks up the cloudy lens, suctions out the cloudy lens and inserts the artificial lens

Effectiveness

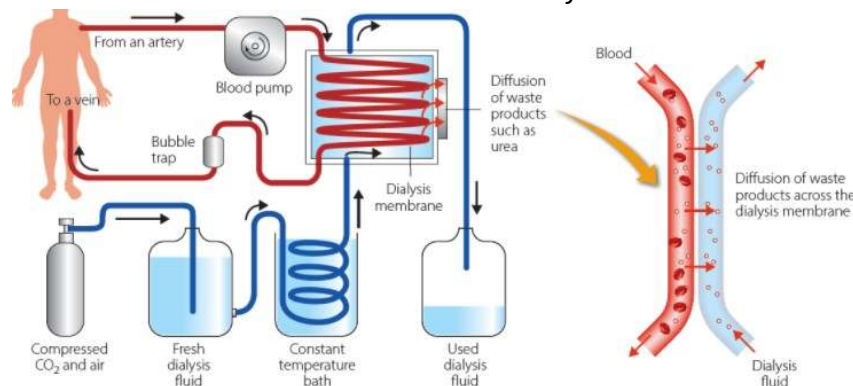
- Takes only 30mins
- Once the artificial lens has been inserted, it is impossible to develop another cataract

- loss of kidney function: dialysis

Kidney Dialysis

Haemodialysis - Blood is pumped out of the body to an artificial kidney machine and returned to the body by tubes that deliver the filtered blood. The machine removes the metabolic wastes that have built up inside the person's blood rather than the kidney.

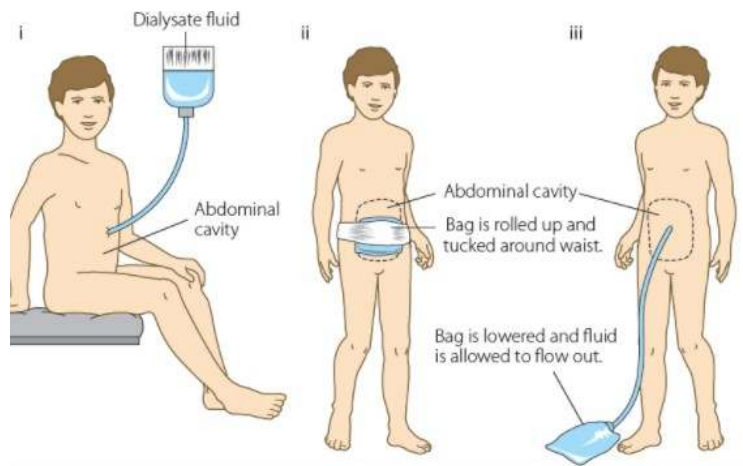
1. Patient is connected to the dialysis machine
2. Machine draws out the patients blood and delivers it to the dialyzer machine
3. Inside the dialyzer machine, blood is pumped through a system of tubes which have semipermeable membranes
4. These tubes are submerged in dialysate (dialysis fluid). Dialysate has the same concentration as blood plasma but without the metabolic wastes
5. Waste materials / toxins move through the semipermeable membrane out of the blood into the dialysate fluid.
6. The blood is now cleaned and ready to be delivered back to the patient.



Peritoneal dialysis - The inside lining of your own belly acts as a natural filter. Wastes are taken out by means of a cleansing fluid called dialysate, which is washed in and out of your belly in cycles. (happens in the peritoneal cavity)

1. A surgical hollow tube is placed in the lower abdomen
2. Dialysate fluid is pumped into the abdominal cavity through a tube coming from a vertically hanging bag
3. The bag is rolled up and tucked around the waist. The fluid is left in the abdominal cavity for the diffusion of waste products to take place

4. Once the process is completed, the bag is lowered and the fluid, which contains the waste material, is allowed to flow out



	Advantages	Limitations
Haemodialysis	- For those who cannot do peritoneal dialysis treatment by themselves (e.g. elderly, visually impaired) - Most hospitals offer haemodialysis and therefore is accessible to most areas	- Required to visit dialysis clinic for treatment - Must arrange dialysis facilities when travelling - Diet and water intake is restricted - Requires repeated large blocks of time (3-4hr sessions) impacting quality of life - Limits patients mobility and is expensive - Substances such as sodium phosphate and potassium ions do not diffuse quickly enough and therefore may accumulate in the blood - Patients must stick to a strict diet to therefore prevent build up of these substances - Must be cycled multiple times as diffusion = equilibrium, therefore each cycle only 50% of waste is removed
Peritoneal	- For those with at least some kidney function - Can be done at home - Equipment is portable - Fewer diet restrictions	- Must be carried out everyday - Only a limited amount of waste is removed at a time - Risk of infection - Can reduce protein levels -> low energy, malnutrition
Both	- Allows the removal of metabolic waste when the kidney no longer functions - Provides a long-term solution and gives patients time when waiting for kidney transplants	- Works only by passive transport and therefore not all wastes can be filtered, as they will only move into the dialysate until equilibrium is reached across the membrane - Neither cure the disease - More expensive in the long run compared to a kidney transplant

How to evaluate a source:

- Is it advertising anything (bias)
- Was it peer reviewed
- Is it up to date
- Is it relevant to research
- Are citations provided

Reliability

- Is it written by a reliable source (Phd, doctor)
- Does it match with other sources of data
- Reputable research hub / organisation